

Introduction

Validating an antenna's performance often requires taking measurements in an anechoic chamber, a controlled environment designed to eliminate external electromagnetic interference and internal wave reflections. In our setup, a vector network analyzer (VNA) measures S-parameters to characterize the antenna under test (AUT), providing insights into the properties such as gain, reflection coefficient, and impedance. To streamline this measurement process, we developed a GUI and data processing tool for the VNA which enables efficient analysis and improves accessibility of measurement data for post-processing. Building upon an existing framework, we have introduced new measurement functions, expanded GUI capabilities to enable more dynamic user input, added interactive graphing abilities, and reorganized data storage methods. Combined, the resulting tool supports automated data acquisition, visualization of S-parameter data, and streamlined post-processing capabilities. These improvements help reduce manual tasks, minimize errors, and improve experiment reproducibility.

Anechoic Chamber Set Up

Inside the chamber we a dual antenna set up with a standard gain horn (SGH) and an antenna under test (AUT). From this set up, we can measure the **gain** and **return loss** of the AUT.

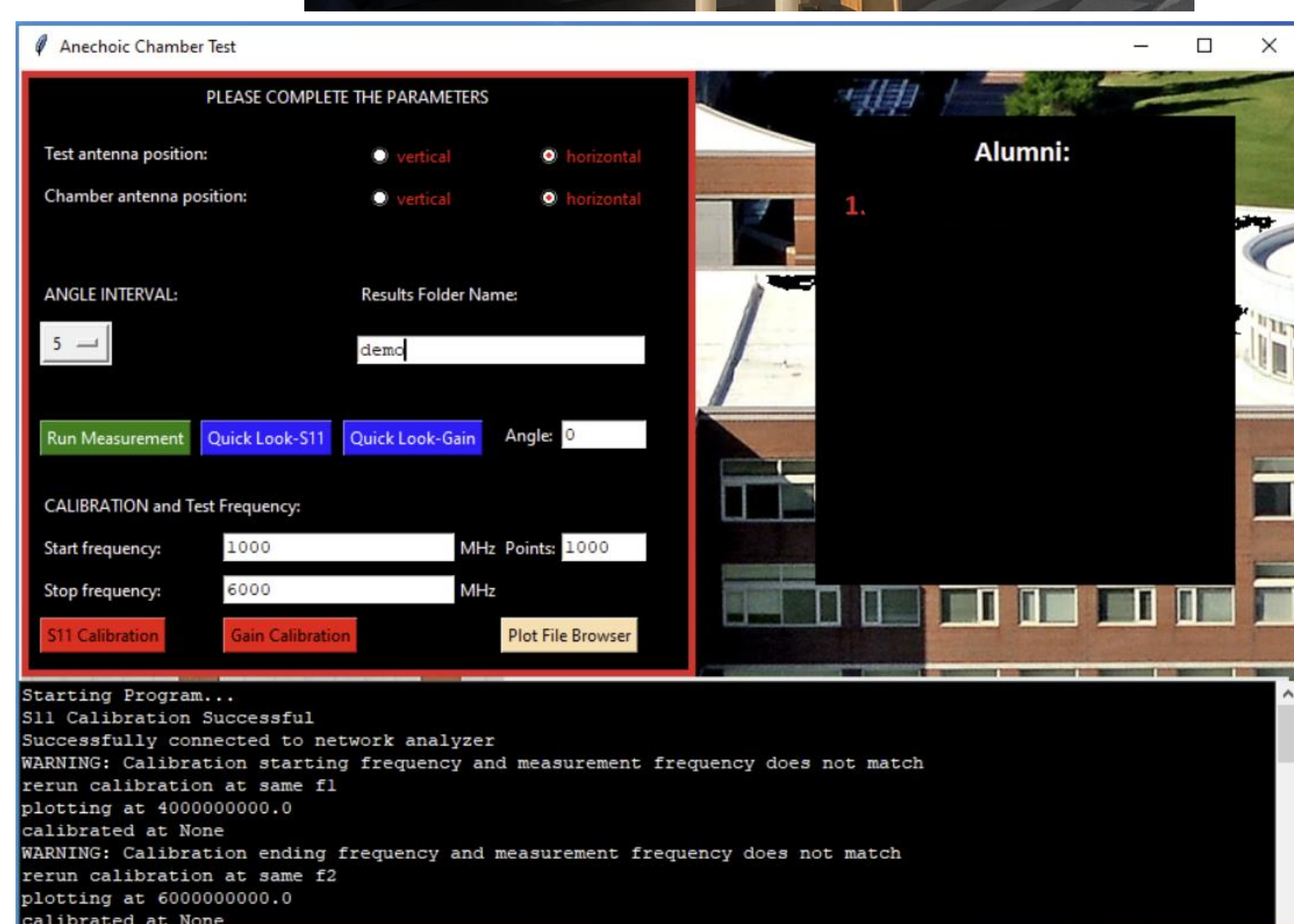
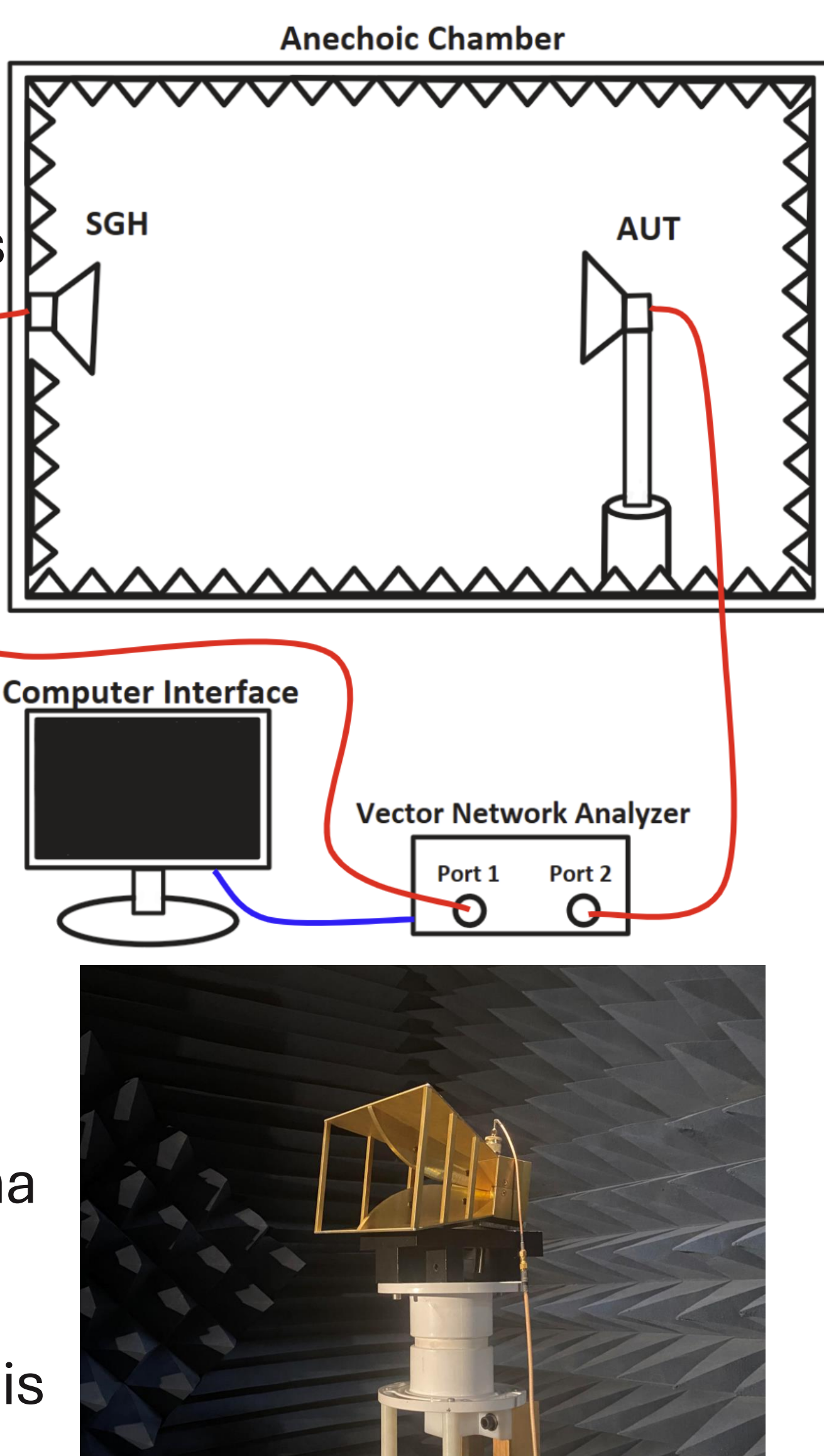
- The **reflection coefficient** tells us how much signal is reflected back over certain frequencies. In other words, it tells us how well matched an antenna's impedance is to a system and how effectively an antenna can receive or transmit a signal. This is found by sending a frequency sweep to port 2 and how much of reflects back to port 2.
- From this, we can find the **return loss** of the antenna by:

$$\text{Return Loss} = -20 \log_{10}(\text{reflection coeff.})$$

- The **gain** of an antenna is how much power is radiated in a certain direction. We can calculate the gain of the antenna measuring the power of each antenna. This can be done by switching which is receiving/transmitting. We can use this plus the known gain of the SGH to calculate the gain of the AUT.

$$G_{\text{AUT}} = G_{\text{SGH}} + (P_{\text{AUT}} - P_{\text{SGH}})$$

These S-parameters help us characterize an antenna's efficiency and radiation pattern. In our final study, we graph and analyze these parameters to compare to its data sheet specs.



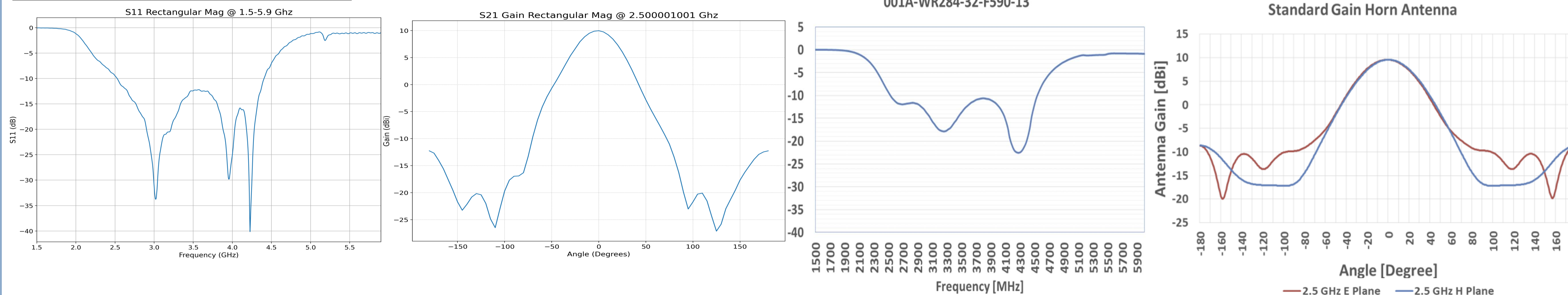
Comparing Chamber Measurements to Data Sheets

To test the accuracy of the chamber, we built a few of antennas using the Anten'it Research Kit to compare our measurements to their data sheets.

Our Measurements

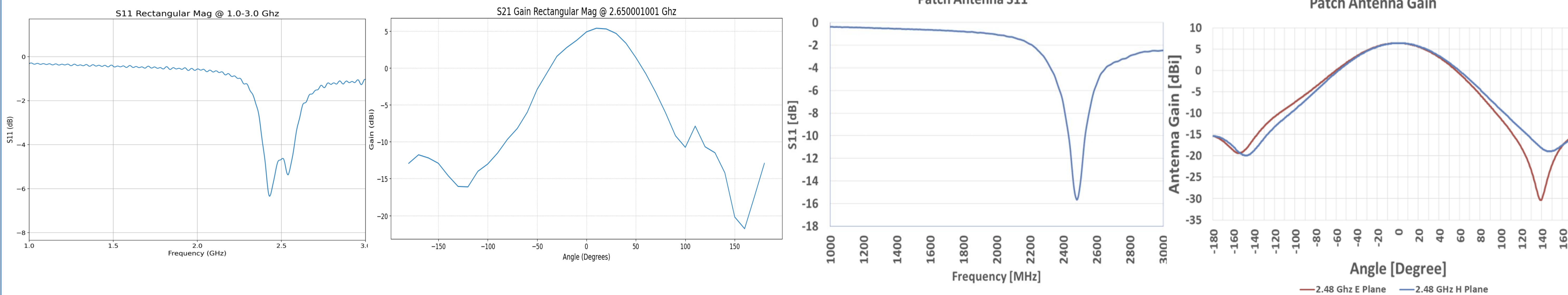
Data Sheet

Standard Gain Horn Antenna



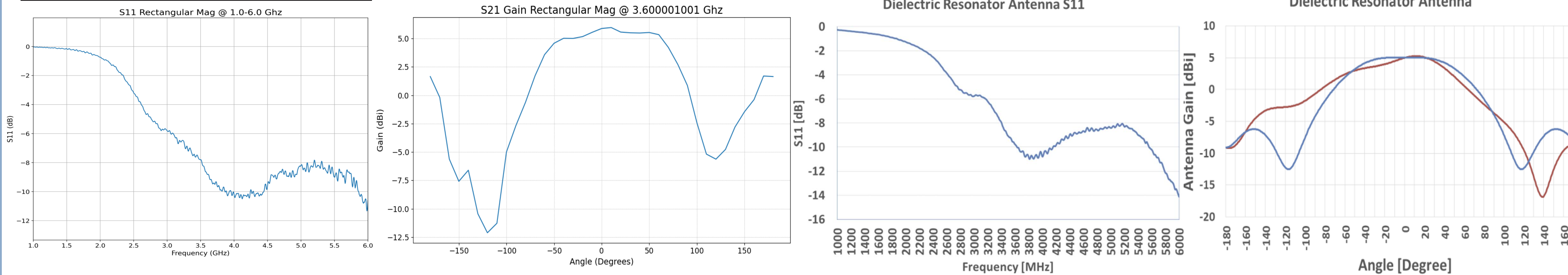
In both return loss graphs, we see nearly identical return loss over its operating frequency. In the 2.5 GHz gain graphs, we see the see 10 dBi at zero degrees and comparable pattern over all angles.

Patch Antenna



In both return loss graphs, we see a spike at roughly the same 2.45 GHz, but with different S11 dips. This is likely due to slight fabrication variances and real world measurement imperfections. Despite this, we were able to get similar gain measurements, but at a higher frequency of 2.65 GHz.

Dielectric Resonator Antenna



In both return loss graphs, we see nearly identical results over its operating frequency. In the 3.6 GHz gain graphs, we see comparable results, but with a larger dip on opposite sides. This may be due to orientation differences.

Future Work

- Add normalization process for live graphing
- Further test the error warnings for S21 calibration processes
- For S21 Quick Looks, change x axis labels to use frequencies from measurement data rather than calibration data

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